# FMO

# Time Value Of Money

# The Interest Rate

Which would you prefer – $10,000 today or $10,000 in 5 years?

Obviously, $10,000 today.

You already recognize that there is *TIME VALUE TO MONEY!!*

# The Interest Rate

TIME allows you the *opportunity* to postpone consumption and earn INTEREST.

# The Interest Rate

- **Simple Interest**

  **Interest paid (earned) on only the original amount, or principal, borrowed (lent).**

- Compound Interest

Interest paid (earned) on any previous interest earned, as well as on the principal borrowed (lent).

# The Interest Rate

*Formula*

$$SI = P_0(i)(n)$$

SI: Simple Interest

$P_0$: Deposit today (t=0)

i: Interest Rate per Period

n: Number of Time Periods

# The Interest Rate

•Assume that you deposit $1,000 in an account earning 7% simple interest for 2 years. *What is the accumulated interest at the end of the 2nd year?*

- $SI = P_0(i)(n)$

$$= \$1{,}000(0.07)(2)$$

$$= \$140$$

# The Future Value

- What is the Future Value (FV) of the deposit?

$$FV = P_0 + SI$$

$$= \$1{,}000 + \$140$$

$$= \$1{,}140$$

- *Future Value* is the value at some future time of a present amount of money, or a series of payments, evaluated at a given interest rate.

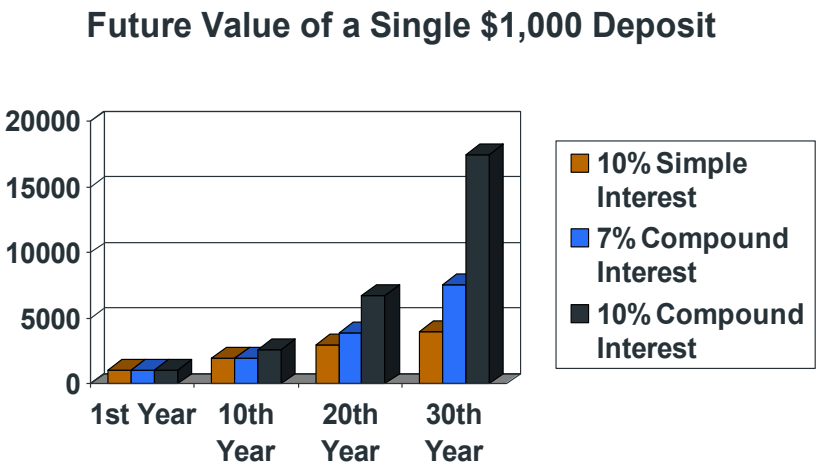
# The Present Value

•What is the Present Value (PV) of the previous problem?

*The Present Value is simply the $1,000 you originally deposited. That is the value today!*

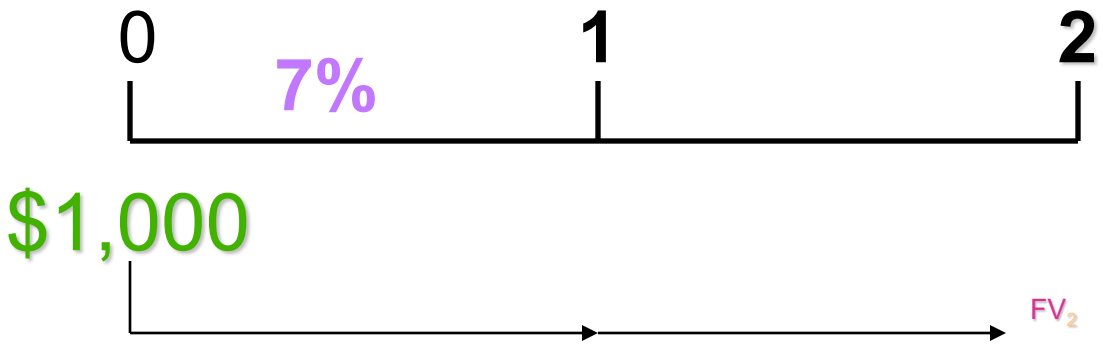
- *Present Value* is the current value of a future amount of money, or a series of payments, evaluated at a given interest rate.

# Compound Interest



# Future Value

Assume that you deposit $1,000 at a compound interest rate of 7% for 2 years.



# Future Value

$$FV_1 = P_0 (1 + i)^1 = \$1{,}000 (1.07)$$
$$= \$1{,}070$$

*Compound Interest*

You earned $70 interest on your $1,000 deposit over the first year. This is the same amount of interest you would earn under simple interest.

# Future Value

$FV_1 = P_0 (1 + i)^1 = \$1{,}000 (1.07)$

$= \$1{,}070$

$FV_2 = FV_1 (1 + i)^1$

$= P_0 (1 + i)(1 + i) =$

$\$1{,}000(1.07)(1.07) = P_0 (1 + i)^2$

$= \$1{,}000(1.07)^2$

$= \$1{,}144.90$

You earned an *EXTRA $4.90* in Year 2 with compound over simple interest.

# Future Value

$$FV_1 = P_0(1 + i)^1$$
$$FV_2 = P_0(1 + i)^2$$

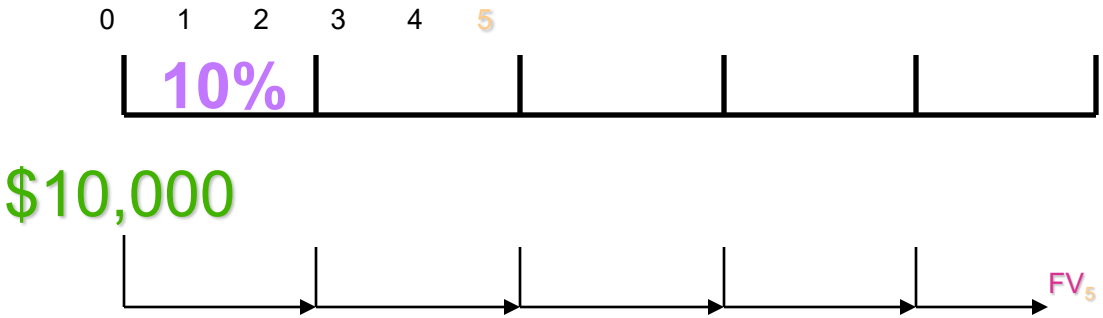
General Future Value Formula:

$$FV_n = P_0 (1 + i)^n$$

or $FV_n = P_0 (FVIF_{i,n})$ – *See Table I*

# Case Example

Julie Miller wants to know how large her deposit of $10,000 today will become at a compound annual interest rate of 10% for 5 years.



# Case Example

- **Calculation based on general formula:**

$$FV_n = P_0 (1 + i)^n$$

$$FV_5 = \$10{,}000 (1 + 0.10)^5$$

$$= \$16{,}105.10$$

- Calculation based on Table I:

$$FV_5 = \$10{,}000 (FVIF_{10\%,\ 5})$$

$$= \$10{,}000 (1.611)$$

$$= \$16{,}110$$ [*Due to Rounding*]

# Double the money

Quick! How long does it take to double $5,000 at a compound rate of 12% per year (approx.)?

**Rule of 72**

# Double the money

Quick! How long does it take to double $5,000 at a compound rate of 12% per year (approx.)?

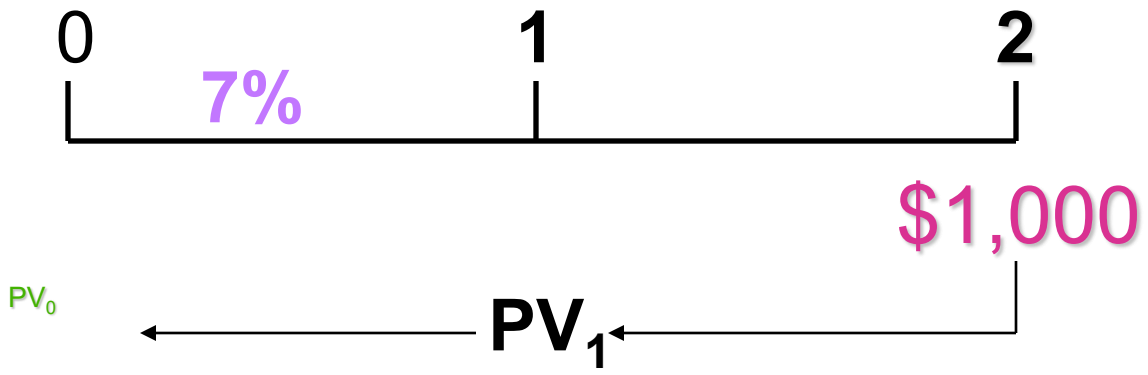
*Approx. Years to Double = 72 / i%*

72 / 12% = *6 Years*

[Actual Time is 6.12 Years]

# Present Value

Assume that you need $1,000 in 2 years. Let's examine the process to determine how much you need to deposit today at a discount rate of 7% compounded annually.



# Present Value

$$PV_0 = FV_1 / (1 + i)^1$$
$$PV_0 = FV_2 / (1 + i)^2$$

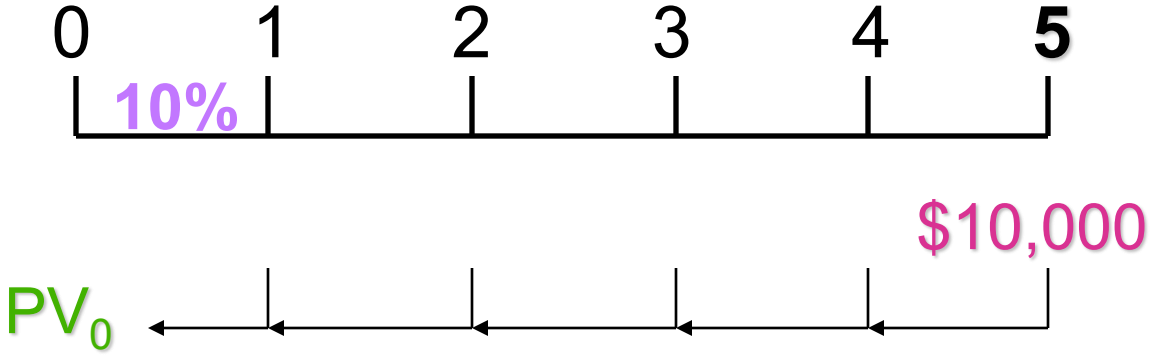
General Present Value Formula:

$$PV_0 = FV_n / (1 + i)^n$$

or $$PV_0 = FV_n (PVIF_{i,n})$$

# Present Value

Julie Miller wants to know how large of a deposit to make so that the money will grow to $10,000 in 5 years at a discount rate of 10%.



# Present Value

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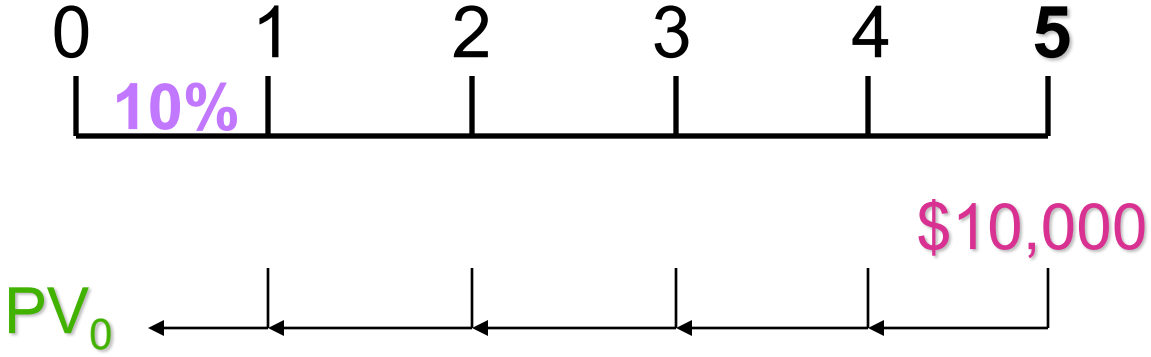


# NPV and IRR

**Net present value (NPV):** the sum of the present values of all cash inflows minus the sum of the present values of all cash outflows.

The **internal rate of return (IRR):** (1) the discount rate that equates the sum of the present values of all cash inflows to the sum of the present values of all cash outflows; (2) the discount rate that sets the **net present value** equal to zero.

The **internal rate of return** measures the investment yield.

# NPV and IRR

**NPV: The sum of the present values of a project's cash inflows and outflows.**

**Choosing the appropriate discount rate accounts for risk.**

$$NPV = -CF_0 + \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \ldots + \frac{CF_N}{(1+r)^N}$$

**Accept projects if NPV > 0.**

# NPV and IRR

**IRR: the discount rate that results in a zero NPV for a project.**

$$NPV = 0 = -CF_0 + \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \ldots + \frac{CF_N}{(1+r)^N}$$

**The IRR decision rule for an investing project is:**

- If IRR is *greater than* the cost of capital, *accept* the project.
- If IRR is *less than* the cost of capital, *reject* the project.

# NPV and IRR

***Gladstone ltd is considering two mutually exclusive projects, A and B, each of which requires an initial outlay of Rs 50 lakh. Cost of capital of the company is 12 %.The expected cash inflows of two projects are:***

| Year | Project- A (in lakh) | Project-B (in lakh) |
|---|---|---|
| 1 | 11 | 38 |
| 2 | 19 | 22 |
| 3 | 32 | 18 |
| 4 | 37 | 10 |

# NPV and IRR

NPV:

Project-A: Rs 21.259 lakh

Project-B: Rs 20.633 lakh

IRR rule may disagree with the NPV Rule.

If that is the case **always** go with the NPV Rule

.

# NPV and IRR

A positive NPV means that the project is expected to add value to the firm and will therefore increase the wealth of the owners.

Since our goal is to increase owner wealth, NPV is a direct measure of how well this project will meet our goal.

# NPV and IRR

You are looking at a new project and you have estimated the following cash flows:

Year 0: CF = -165,000
Year 1: CF = 63,120;
Year 2: CF = 70,800;
Year 3: CF = 91,080;

Your required return for assets of this risk is 12%.

$$NPV = 63{,}120/(1.12) + 70{,}800/(1.12)^2 + 91{,}080/(1.12)^3 - 165{,}000 = 12{,}627.42$$

***Do we accept or reject the project?***

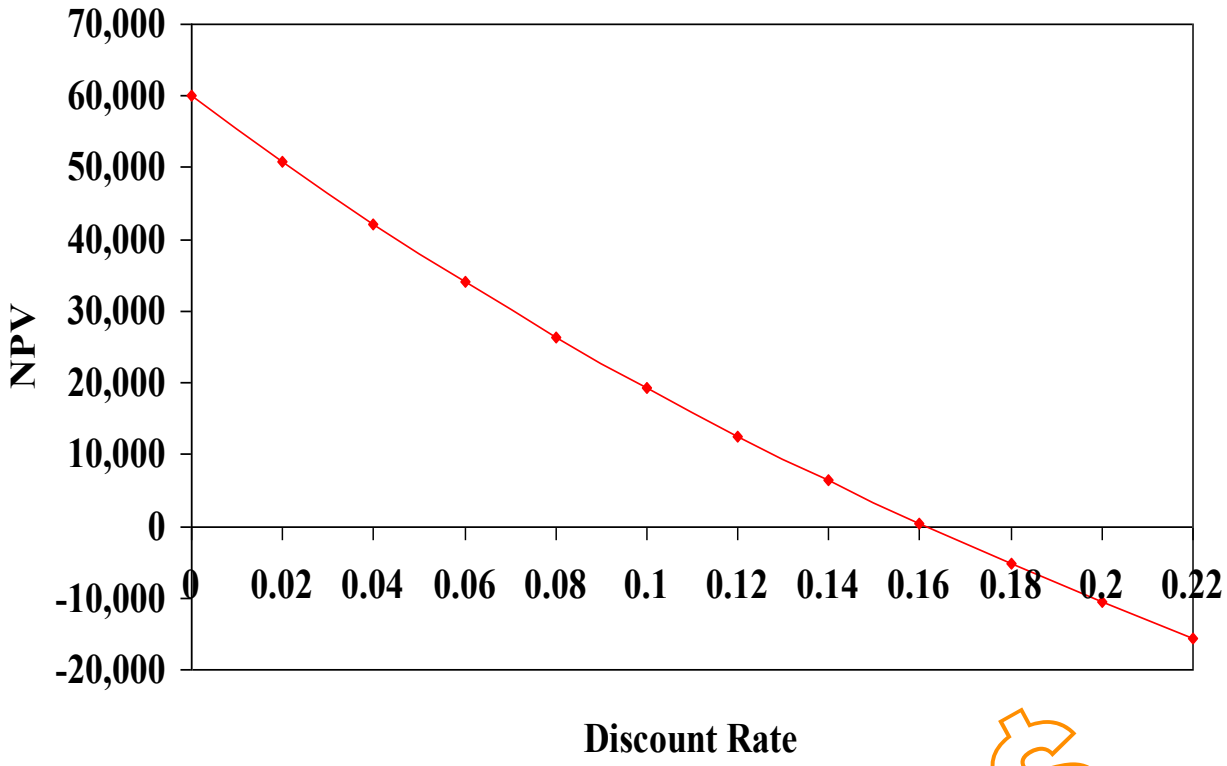
# NPV and IRR

Spreadsheets are an excellent way to compute NPVs, especially when you have to compute the cash flows as well.

Using the NPV function

- ✓ The first component is the required return entered as a decimal
- ✓ The second component is the range of cash flows *beginning with year 1*
- ✓ Subtract the initial investment after computing the NPV

# NPV and IRR

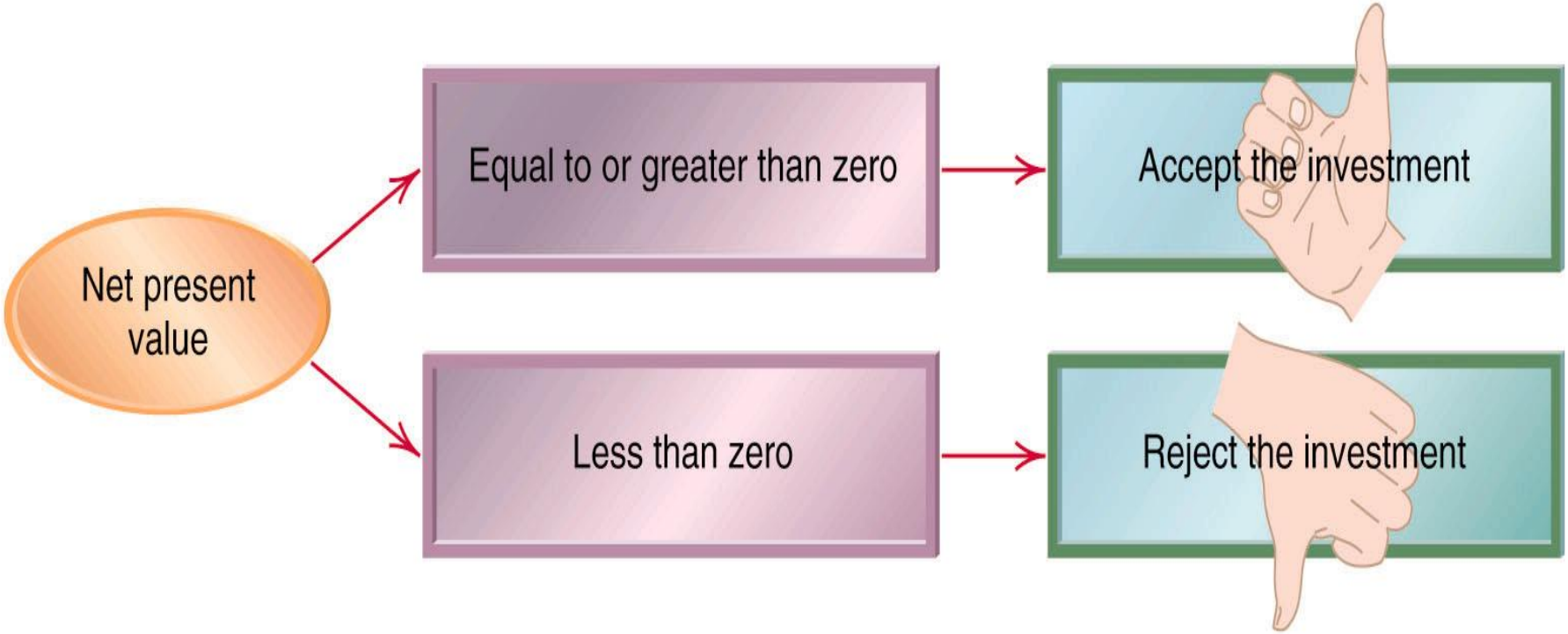

70,000
60,000
50,000
40,000
30,000
20,000
10,000
0
-10,000
-20,000
NPV
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
Discount Rate

# NPV and IRR

You start with the cash flows the same as you did for the NPV
You use the IRR function

- You first enter your range of cash flows, beginning with the initial cash flow
- The default format is a whole percent – you will normally want to increase the decimal places to at least two

# NPV and IRR


Net present value
Equal to or greater than zero
Accept the investment
Less than zero
Reject the investment

# NPV and IRR

| | A | B | C |
|---|---|---|---|
| 1 | Annual RRR (Discount) | 0.15 | |
| 2 | Cash Flow 0 (Cost) | -$200,000.00 | |
| 3 | Cash Flow 1 | $100,000.00 | |
| 4 | Cash Flow 2 | $90,000.00 | |
| 5 | Cash Flow 3 | $70,000.00 | |
| 6 | NPV =Value added to firm if project is undertaken | $1,035.59 | =NPV(B1,B3:B5)+B2 |
| 7 | | | |
| 8 | =NPV(rate, value1, value2, …) + $CF_0$<br>=NPV(i/n, $CF_1$, $CF_2$, …) + $CF_0$ | | |
| 9 | Note: value1 can be the whole range of cash flows | | |

# NPV and IRR

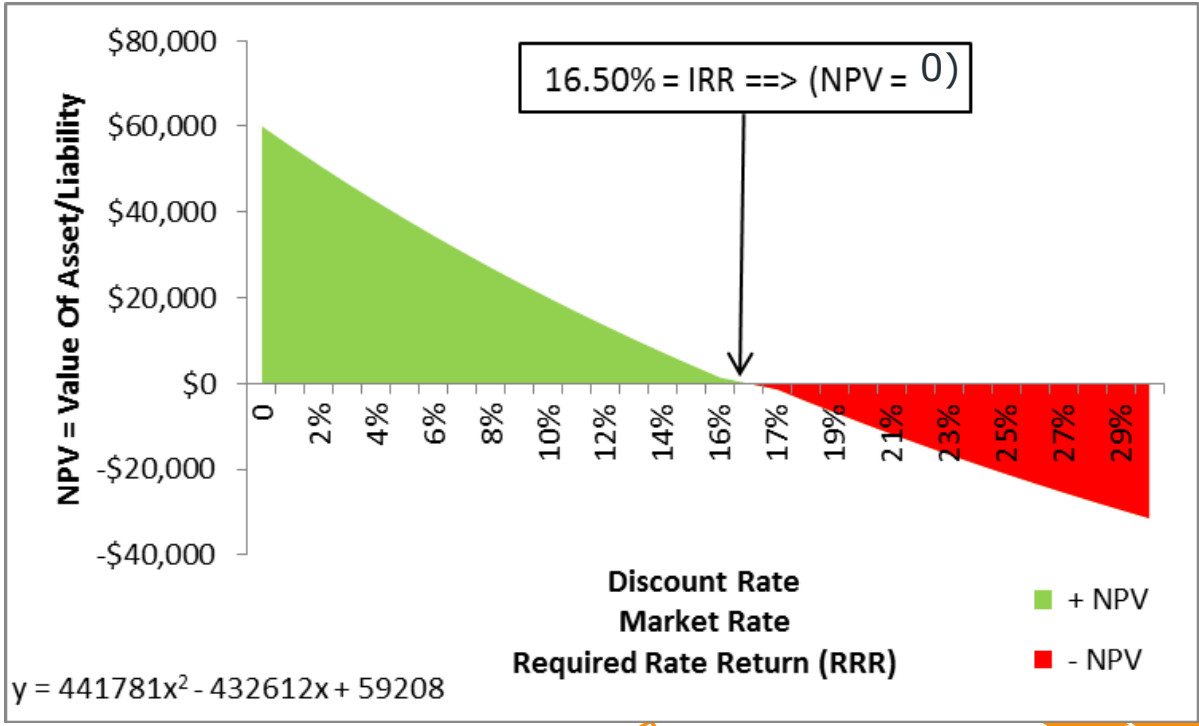
To Understand What IRR means, build a NPV Profile and look for the rate at which NPV = $0

This tells you the rate of return for the cash flows from the project.

# NPV and IRR

| Project Cash Flows | |
|---|---|
| $CF_0$ | -160,000.00 |
| $CF_1$ | 60,000.00 |
| $CF_2$ | 70,000.00 |
| $CF_3$ | 90,000.00 |
| Required Rate Return | 0.15 |

| RRR | NPV |
|---|---|
| 14.00% | 7,241.74 |
| 14.50% | 5,750.17 |
| 15.00% | 4,280.43 |
| 15.50% | 2,832.09 |
| 16.00% | 1,404.73 |
| 16.50% | 0.00 |
| 17.00% | -1,386.63 |
| 17.50% | -2,753.47 |
| 18.00% | -4,100.89 |
| 18.50% | -5,429.27 |
| 19.00% | -6,738.96 |



# NPV and IRR

Formula Inputs are cash flows

| | A | B | C |
|---|---|---|---|
| 1 | Project Cash Flows | | |
| 2 | $CF_0$ | -160,000.00 | |
| 3 | $CF_1$ | 60,000.00 | |
| 4 | $CF_2$ | 70,000.00 | |
| 5 | $CF_3$ | 90,000.00 | |
| 6 | Required Rate Of Return | 0.15 | |
| 7 | IRR = | 0.1649927 | =IRR(B2:B5) |
| 8 | If the Required Return is less than or equal to 16.499%, then accept, otherwise, reject project. | | |
| 9 | Because 15.00% <= 16.50%, we accept the project. | | |
| 10 | When RRR is 16.50%, NPV = 0 | | |

# NPV and IRR

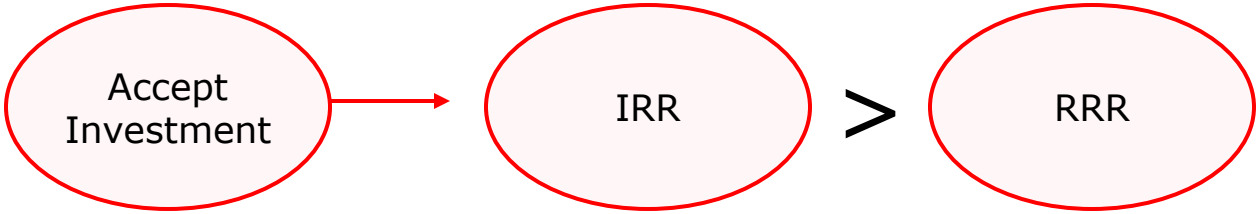
- Knowing a return is intuitively appealing.
- It is a simple way to communicate the value of a project to someone who doesn't know all the estimation details.
- If the IRR is high enough, you may not need to estimate a required return, which is often a difficult task.

In the working world, many people use IRR.

# IRR = Internal Rate of Return =“Break Even Rate”

- **Definition:** Rate that makes the NPV = $0

- **Decision Rule:**




- Most important alternative to NPV.
- It is often used in practice and is intuitively appealing.
- Calculation based entirely on the estimated cash flows and is independent of interest rates found elsewhere
- Formula inputs are cash flows only!

# Net Present Value (NPV) = Discounted Cash Flow Valuation (DCF)

- The process of valuing an investment (project) by discounting its future cash flows
- Decision Rule:
  - NPV > 0 ➔ Accept Project
  - NPV < 0 ➔ Reject Project
  - NPV = 0 ➔ Indifferent (RRR = IRR)

There are no guarantees that our estimates are correct

Create value for stockholder → Search for capital budget projects → That yield positive NPV value added

# DCF -Discounted Cash Flow

The process of valuing an investment (project) by discounting its future cash flows

Decision Rule:

NPV > 0 ➔ Accept Project
NPV < 0 ➔ Reject Project
NPV = 0 ➔ Indifferent (RRR = IRR)

# DCF -Discounted Cash Flow

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Should we launch new product that will be around 5 years that has these cash flows? | | | | |
| 2 | | | | | |
| 3 | Time | All Cash Out | All Cash In | Net Cash Flows | |
| 4 | Time 0 | -100,000.00 | | | |
| 5 | Time 1 | -28,000.00 | 80,000.00 | 52,000.00 | |
| 6 | Time 2 | -16,500.00 | 60,000.00 | 43,500.00 | |
| 7 | Time 3 | -15,000.00 | 40,000.00 | 25,000.00 | |
| 8 | Time 4 | -10,000.00 | 35,000.00 | 25,000.00 | |
| 9 | Time 5 | -20,000.00 | 18,000.00 | -2,000.00 | |
| 10 | | | | | |
| 11 | | | Annual RRR (Discount) | 0.2 | |
| 12 | | | NPV | -738.17 | =NPV(D11,D5:D9)+B4 |
| 13 | | | Answer: | No | |

# Quiz

What does NPV measure?

a) Profitability of an investment

b) Liquidity of a company

c) Return on Investment (ROI)

d) Cash flows over a period of time

# Quiz

Profitability of an investment

# Quiz

What is the IRR of an investment?

a) The rate at which the project breaks even

b) The annualized return earned on an investment

c) The discount rate that makes the NPV of an investment zero

d) The total cash inflow over the life of the investment

# Quiz

The discount rate that makes the NPV of an investment zero

# Quiz

In DCF analysis, which of the following cash flows are discounted?

a) Future cash flows

b) Historical cash flows

c) Present cash flows

d) All of the above

# Quiz

Future cash flows

# Quiz

What happens to the NPV of a project when the discount rate increases?

a) NPV increases

b) NPV decreases

c) NPV remains unchanged

d) NPV becomes negative

# Quiz

NPV decreases